PRODUCT USER MANUAL

For Baltic Sea and North Sea L4 Hourly Sea Surface Temperature

SST_BAL_PHY_SUBSKIN_L4_NRT_010_034

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2022/06/15	All	Initial version	Ioanna Karagali	Copernicus Marine Product Management
2.0	2025/05/28	All	Updated template, addition of VIIRS on NOAA21	Ioanna Karagali	Copernicus Marine Product Management

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a) Structure and semantic of NetCDF maps files..... **Erreur ! Signet non défini.**

GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
BAL	Baltic Sea
SST	Sea Surface Temperature
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
NOAA	National Oceanographic and Atmospheric Administration
OSPO	Office of Satellite and Product Operations
FMI	Finnish Meteorological Institute

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the Product User Manual describing the Copernicus Marine Service SST near-real-time, hourly (level 4) product over the Baltic Sea and North Sea (SST-BAL-PHY-SUBSKIN_L4-NRT-010-034), processed at the Danish Meteorological Institute (DMI). The manual covers how the product was created, content, and which data services are available to access it.

The Copernicus Marine Service Baltic Sea/North Sea L4 Diurnal Sub-skin SST Analysis product is a gap free satellite sub-skin SST analysis created by the DMIOI system (Høyer and She, 2007) using L2P and L3 single-sensor SST data as input. The product is available from November 01 2022 on a regular latitude/longitude grid at 0.02° resolution.
The Copernicus Marine Service Baltic Sea/North Sea L4 Diurnal Sub-skin SST Analysis product provides an estimate of the hourly sub-skin SST with uncertainty estimates, which is the SST including diurnal variability.

The Quality of the product is described in the Quality Information Document (QUID) CMEMS-SST-QUID-010-034

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	SST_BAL_PHY_SUBSKIN_L4_NRT_010_034
Geographical coverage	Baltic Sea and North Sea : 10°W-30°E ; 48°S – 66°N
Variables	• Analysed Sea Surface Temperature • Analysis Error • Sea Ice Fraction
Product Type	Near Real Time - Analysis
Available time series	2022/05/01 to -1 Day
Temporal resolution	Hourly
Update Frequency	Daily
Target delivery time	daily at 07 am UTC
Horizontal resolution	1/50 degree
Number of vertical levels	1 (surface only)
Format	NetCDF4

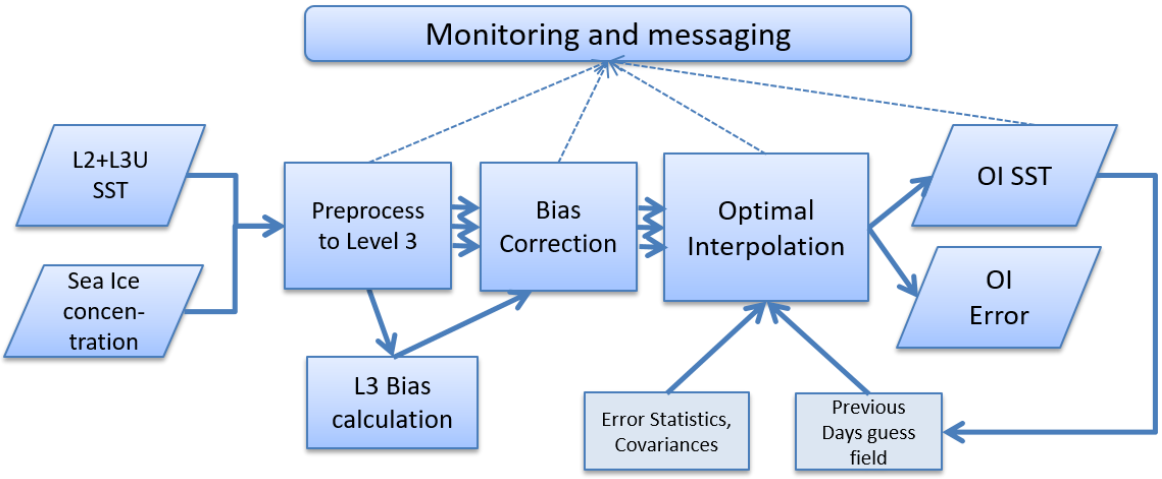
b) Production System Description

The Optimal Interpolation (DMIOI) system is run daily in the operational suite at DMI. It is run at 06:00 UTC and is completed before 09:30 UTC for all hourly files of the previous day. Products are typically available to users starting at 06:30 UTC.

All satellite data valid for a particular day and hour, within 24 hours from the analysis are considered. The input L2/L3U SST data undergo various QC and processing steps to generate separate level 3 products:

- Only satellite SST satellite data, which have a quality flag of 4 and higher are accepted.
- Single-sensor hourly files are bias-corrected to ensure consistency between all input SST.
- Hourly anomalies, dSST, from the SST foundation temperature of the same day are estimated as SST_{hour} minus $SST_{foundation}$. The OI is then applied to the hourly anomaly field, to produce gap-free hourly dSST and the $SST_{foundation}$ is subsequently added to the hourly dSST.

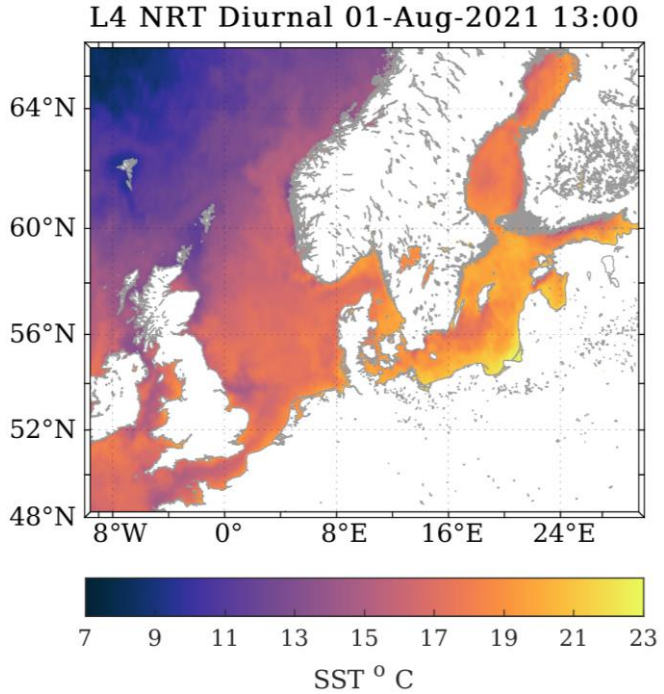
The main SST analysis uses a persistence approach based on the use of the previous analysis field as a first guess. The collated L3 (L3S) SST hourly observations from the current day/hour are interpreted as anomalies with respect to the first guess field and local statistics have been included to estimate the optimal estimates SST values for the final field (Høyer and She, 2007).



The system running at DMI.

North Sea/Baltic Sea (10°W-30°E; 48°S – 66°N): 1/50°, regular grid, 2001 x 901, hourly

This product is regional, defined on a standard collocated grid at 1/50 degree (approx. 2.2 km) and 24 fields are produced per day, one for each hour.



Vertical grid	1 vertical level
System Name	DMIOI
Model version	
Horizontal Resolution	1/50°
Vertical Resolution	1 Z level
Interpolation method	Only horizontal interpolation : optimal
Assimilated observations	SST L2P satellite data in GHRSST compliant format, i.e. GHRSST GDS V2.0 [RD-1], from InfraRed sensors: • VIIRS (SUOMI/NPP, NOAA20, NOAA21) • SLSTR on Sentinel-3A/B • AVHRR on MetOp-A/B/C • SEVIRI on MSG See quality information document for full details. Sea-ice concentration data: Baltic high resolution sea ice concentration from the Copernicus Marine Service SI TAC (SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004)

c) Dataset details

Product Lines	SST_BAL_PHY_SUBSKIN_L4_NRT_010_034

Geographical coverage	Baltic Sea/North Sea
Datasets: cmems-obs-sst-bal-phy-subskin-nrt-l4_P1H-m	
Product Type	Analysis
Update frequency	Daily
Available time series	2022/11/01 - up to T-1day
Target delivery time	Daily starting at 06:30am UTC
Temporal resolution	Hourly mean

Dataset : cmems_obs-sst_bal_phy-subskin_nrt_l4_P1H-m contains the 2D hourly fields: sea surface temperature, analysis error, sea ice fraction, mask
analysed_sst [K] analysed sea surface temperature sea_surface_subskin_temperature
analysis_error [K] estimated error standard deviation of analysed_sst sea_surface_subskin_temperature standard_error
sea_ice_fraction [ ] sea ice area fraction sea_ice_area_fraction
mask [ ] land sea ice lake bit mask

d) Processing information

Product disseminating schedule :

Type of product (forecast/ hindcast)	Forecast run (base time)	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
Analysis	00 UTC	0 to 24	06:30am -> 09:30am	Every day

e) Catalogue updating

Every day hourly L4 SST analyses of the previous day are delivered.

f) Time averaging

Dataset frequency	Time averaging	Forecast lead time
Hourly-mean	Each hourly SST analysis is based on aggregated input data from the analysis day and hour. The previous day's analysis is used as a first guess field for the OI processing system.	N/A

g) Product use and impact

For applications like daily SST monitoring and up-to-date analysis, there are several reasons to choose the hourly L4 product specifically developed for the Baltic Sea and North Sea. Some of the advantages are listed below:

- Increased spatial resolution gives a better representation of fronts and near coastal effects.
- Regional, high resolution land mask gives a better representation of small islands and complex coastal regions.
- The high resolution 1 km sea ice product derived from ice charts is significantly better in this coastal region than the global Microwave-based products.
- Regional Interpolation statistics improves the results in case of missing data.
- The hourly data availability makes it ideal for monitoring diurnal variability of SST especially during the spring and summer periods, when this signal is most pronounced in the Baltic Sea and can also be associated with fast-developing, harmful algal-blooms. Moreover, in this region of high offshore renewable resource installations, hourly SSTs can help modeling activities of the wind resources.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1 FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_obs-sst_bal_phy-subskin_nrt_l4_PT1H-m	<yyyymmddhh>0000-DMI-L4_GHRSST-SSTsubskin-DMI_OI-NSEABALTIC-v02.0-fv01.0.nc	0.011

4.2 STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5) REFERENCES

Quality Information Document
https://documentation.marine.copernicus.eu/QUID/CMEMS-SST-QUID-010-034.pdf

Høyer, J. L. and She, J., Optimal interpolation of sea surface temperature for the North Sea and Baltic Sea, J. Mar. Sys., Vol 65, 1-4, pp. 176-189, 2007.

6) ANNEX

For dataset cmems_obs-sst_bal_phy-subskin_nrt_l4_PT1H-m

```
netcdf \20250430230000-DMI-L4_GHRSST-SSTsubskin-DMI_OI-NSEABALTIC-v02.0-fv01.0
{
dimensions:
        lon = 2001 ;
        lat = 901 ;
        time = 1 ;
variables:
        float lon(lon) ;
                lon:standard_name = "longitude" ;
                lon:long_name = "longitude" ;
                lon:units = "degrees_east" ;
                lon:axis = "X" ;
                lon:valid_min = -180.f ;
                lon:valid_max = 180.f ;
                lon:comment = "geographical coordinates, WGS84 projection" ;
        float lat(lat) ;
                lat:standard_name = "latitude" ;
                lat:long_name = "latitude" ;
                lat:units = "degrees_north" ;
                lat:axis = "Y" ;
                lat:valid_min = -90.f ;
                lat:valid_max = 90.f ;
                lat:comment = "geographical coordinates, WGS84 projection" ;
        int time(time) ;
                time:standard_name = "time" ;
                time:long_name = "reference time of sst field" ;
                time:units = "seconds since 1981-01-01 00:00:00" ;
                time:axis = "T" ;
                time:comment = "Nominal time of Level 4 analysis" ;
        short analysed_sst(time, lat, lon) ;
                analysed_sst:standard_name = "sea_surface_subskin_temperature" ;
                analysed_sst:long_name = "analysed hourly sea surface temperature" ;
                analysed_sst:units = "kelvin" ;
                analysed_sst:scale_factor = 0.01f ;
                analysed_sst:add_offset = 273.15f ;
                analysed_sst:_FillValue = -32768s ;
                analysed_sst:comment = "" ;
                analysed_sst:source = "OSI-SAF and GHRSST L2P/L3 satellite SST products"
;
                analysed_sst:valid_min = -300s ;
                analysed_sst:valid_max = 4500s ;
        short analysis_error(time, lat, lon) ;
                analysis_error:long_name = "estimated error standard deviation of
analysed_sst" ;
```

```
                analysis_error:units = "kelvin" ;
                analysis_error:scale_factor = 0.01f ;
                analysis_error:add_offset = 0.f ;
                analysis_error:_FillValue = -32768s ;
                analysis_error:comment = "" ;
                analysis_error:valid_min = 0s ;
                analysis_error:valid_max = 32767s ;
        byte mask(time, lat, lon) ;
                mask:long_name = "land sea ice lake bit mask" ;
                mask:_FillValue = -128b ;
                mask:comment = "Mask can be used to further filter the data" ;
                mask:flag_meanings = "water land optional_lake_surface sea_ice
optional_river_surface" ;
                mask:source = "NAVOCEANO_landmask_v1.0, Myocean_HR_ice" ;
                mask:flag_masks = 1b, 2b, 4b, 8b, 16b ;
                mask:valid_min = 1b ;
                mask:valid_max = 31b ;
        byte sea_ice_fraction(time, lat, lon) ;
                sea_ice_fraction:standard_name = "sea_ice_area_fraction" ;
                sea_ice_fraction:long_name = "sea ice area fraction" ;
                sea_ice_fraction:units = "1" ;
                sea_ice_fraction:scale_factor = 0.01f ;
                sea_ice_fraction:add_offset = 0.f ;
                sea_ice_fraction:_FillValue = -128b ;
                sea_ice_fraction:comment = "" ;
                sea_ice_fraction:source = "MyOcean HR icemask" ;
                sea_ice_fraction:valid_min = 0b ;
                sea_ice_fraction:valid_max = 100b ;

// global attributes:
                :Conventions = "CF-1.4, Unidata Observation Dataset v1.0" ;
                :title = "North Sea/Baltic Sea SST L4 Diurnal Sub-skin Sea Surface
Temperature Analysis (SST_BAL_PHY_SUBSKIN_L4_NRT_010_034, cmems_obs-
sst_bal_phy-subskin_nrt_l4_PT1H-m)" ;
                :summary = "DMI L4 Diurnal Sub-skin Sea Surface Temperature Analysis
created using multi-sensor satellite SST observations" ;
                :references = "Høyer, J. L. and She, J., Optimal interpolation of sea
surface temperature for the North Sea and Baltic Sea,  J. Mar. Sys., Vol 65, 1-4, pp. 176-
189, 2007" ;
                :institution = "Danish Meteorological Institute, DMI" ;
                :history = "Version 1.0" ;
                :license = "these data are available free of charge under the Copernicus
Marine Service data policy" ;
                :id = "DMI-L4_GHRSST-SSTsubskin-NSEABALTIC-DMI_OI" ;
                :naming_authority = "org.ghrsst" ;
                :product_version = "Version 1.0" ;
                :uuid = "428965f4-053d-4827-8fd5-5c4693cb7548" ;
                :gds_version_id = "2.0" ;
                :netcdf_version_id = "netcdf-4.4.1.1.6" ;
                :date_created = "20250501T091204Z" ;
```

:file_quality_level = 3 ;
:spatial_resolution = " 0.02 degrees" ;
:start_time = "20250430T000000Z" ;
:time_coverage_start = "20250430T000000Z" ;
:stop_time = "20250501T000000Z" ;
:time_coverage_end = "20250501T000000Z" ;
:westernmost_longitude = -10.f ;
:easternmost_longitude = 30.f ;
:southernmost_latitude = 48.f ;
:northernmost_latitude = 66.f ;
:source = "VIIRS_SST_NPP_NAR-OSISAF-L3C, SEVIRI_SST-OSISAF-L3C, EUR-L2P-AVHRR_METOP_B, CODA-EUMETSAT-SLSTR_S3A/B, VIIRS_NPP_OSPO, VIIRS_N20_OSPO, MyOcean_HR_ICE" ;
:platform = "NOAA-20, NOAA-21, MetOpB, MSG2, SUOMI NPP, Sentinel 3 A/B" ;
:sensor = "AVHRR, SEVIRI, VIIRS, SLSTR" ;
:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
:metadata_link = "http://data.nodc.noaa.gov/NESDIS_DataCenters/metadata/index.html" ;
:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;
:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;
:standard_name_vocabulary = "NetCDF Climate and forecast (CF) Metadata Convention" ;
:geospatial_lat_units = "degrees north" ;
:geospatial_lon_units = "degrees east" ;
:geospatial_lat_resolution = 0.02f ;
:geospatial_lon_resolution = 0.02f ;
:acknowledgment = "Please acknowledge the use of these data with the following statement: These data were provided by DMI and Copernicus Marine Service" ;
:creator_name = "Ioanna Karagali" ;
:creator_email = "ika@dmi.dk" ;
:creator_url = "http://ocean.dmi.dk/satellite/" ;
:project = "Copernicus CMEMS SST TAC project" ;
:publisher_name = "Copernicus Marine Service SST TAC and DMI" ;
:publisher_url = "https://marine.copernicus.eu/" ;
:publisher_email = "servicedesk.cmems@mercator-ocean.eu" ;
:processing_level = "L4" ;
:cdm_data_type = "grid" ;
:comment = "IN NO EVENT SHALL DMI OR ITS REPRESENTATIVES BE LIABLE FOR ANY DAMAGES WHATSOEVER INCLUDING, WITHOUT LIMITATION, SPECIAL, INDIRECT, INCIDENTAL OR CONSEQUENTIAL DAMAGES OR DAMAGES FOR LOSS OF BUSINESS PROFITS OR SAVINGS, BUSINESS INTERRUPTION, LOSS OF BUSINESS INFORMATION OR OTHER PECUNIARY LOSS ARISING OUT OF THE USE OF OR THE BUSINESS INTERRUPTION, LOSS OF BUSINESS INFORMATION OR OTHER PECUNIARY LOSS ARISING OUT OF THE USE OF OR THE INABILITY TO USE THIS DMI PRODUCT, EVEN IF DMI HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THIS LIMITATION SHALL APPLY TO CLAIMS OF PERSONAL INJURY TO THE EXTENT PERMITTED BY LAW. SOME COUNTRIES OR STATES DO NOT ALLOW

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}